Biochemistry Cumulative Examination
April 1, 2011

Answer the following questions about fatty acid metabolism.

1. A key part of fatty acid metabolisms is so-called "ketone bodies." Name the two principal ketone bodies. Tell what molecule(s) they are formed from. Explain why these two ketone bodies are formed. In other words, what metabolic purpose do they serve and what metabolic fate awaits them? 25 points

2. A third ketone body, acetone, is formed as an inadvertent byproduct. Explain how this happens. 10 points

3. What advantages do fatty acids have as a store of energy relative to carbohydrates? What disadvantages? 25 points

4. Free fatty acids are seldom found. To what enzymatic cofactor are fatty acids commonly bound before being oxidized for energy? 10 points

5. What purpose does unsaturation of fatty acids serve? 15 points

6. How does the degradation of fatty acids with odd numbers of carbon differ from that of fatty acids with even numbers of carbon? 15 points